Congenital Heart Defects				
Disorder	**Definition**	**Symptoms**	**Diagnosis**	**Treatment**
Atrial Septal Defect **Acyanotic**	Opening in the atrial septum allowing blood to shunt from the left to right atria.	s/s of L-sided HF: increased WOB (tachypnea, dyspnea, crackles), poor weight gain	Echocardiogram, Chest X-ray, ECG	No tx if asymptomatic management of symptoms: furosemide surgical closure: occlusion via cardiac cath
Ventricular Septal Defect **Acyanotic**	Opening in the ventricular septum allowing blood flow from the left to the right ventricle	s/s of L-sided HF: increased WOB (tachypnea, dyspnea, crackles), poor weight gain	Echocardiogram, Chest X-ray, ECG	No treatment if asymptomatic Medication symptom management: furosemide Some types closed via cardiac cath Others closed through a median sternotomy
Patent Ductus Arteriosus (PDA) **Acyanotic**	Opening allowing blood flow from the aorta to the pulmonary artery.	s/s of L-sided HF: increased WOB (tachypnea, dyspnea, crackles), poor weight gain	Echocardiogram, Chest X-ray, ECG	Medicinal closure: Indomethacin (prostaglandin inhibitor) Additional medication management: furosemide Surgical closure (PDA ligation or occlusion via cardiac cath
Coarctation of the Aorta (COA) **Can be cyanotic or acyanotic based on severity**	Localized narrowing of the aorta at the ductus arteriosus, restricting systemic blood flow (specifically to lower extremities)	◦ Upper extremities: Elevated BP (commonly in right arm), bounding pulses Lower extremities: Hypotension. weak/absent pulses, cool skin	Echocardiogram, Chest X-ray, ECG	Surgical repair-balloon angioplasty (repair may be delayed based on severity)

*** Symptoms and treatment vary based on severity of the defect

***Blood always takes the path of least resistance (goes from high pressure to low pressure)

Tricuspid Atresia **Cyanotic**	Failure of tricuspid valve to develop= no communication between RA and RV (shunting from right to left)	Cyanosis Chronic hypoxemia in older children	Echocardiogram, Chest X-ray, ECG	Surgical repair Prior to surgery: continuous infusion of prostaglandin E1- keep the PDA open!
Tetrology of Fallot (TOF) **Cyanotic**	Four defects: VSD, pulmonary stenosis, overriding aorta, R ventricular hypertrophy.	Cyanosis Hypercyanotic “Tet” spells	Echocardiogram, Chest X-ray, ECG	Surgical repair Prior to surgery: continuous infusion of Prostaglandin E1- keep PDA open!
Transposition of the Great Arteries (TGA) **Cyanotic**	Pulmonary artery and aorta switched positions, causing systemic circulation of oxygen-poor blood (no communication between right and left side)	Profound cyanosis Upper extremities SpO2 < lower extremities	Echocardiogram, Chest X-Ray, ECG	Surgical repair Prior to surgery: continuous infusion of Prostaglandin E1- keep PDA open!
Acquired Heart Disease				
Kawasaki Disease	Generalized vasculitis of all blood vessels and coronary arteries	Initial: prolonged fevers unrelieved by antipyretics REDNESS (eyes, mucous membranes, strawberry tongue) Chapping/peeling to lips/hands/feet Cervical lymphadenopathy Cardiac: dysrhythmias, myocarditis	Clinical evaluation, Echocardiogram, Chest X-Ray ECG Laboratory data	IV Gamma Globulin (IVIG), ASA, Hospitalization during acute phase
Rheumatic Fever	Autoimmune response due to group A beta-hemolytic strep (GABHS) leading to inflammatory lesions in the heart, blood vessels, brain, and joints	Jones criteria: carditis (most serious), polyarthritis, chorea, macular rash, subcutaneous nodules, fever, arthralgia	Clinical evaluation Echocardiogram Chest X-Ray Laboratory data	◦ Penicillin (treat GABHS) ◦ NSAIDs (reduce inflammation/joint pain) ◦ Phenobarbital and diazepam (to reduce chorea)